

Vibrotactile Invisible Presence: Conveying Remote Presence through Moving Vibrotactile Footstep Cues on a Haptic Floor

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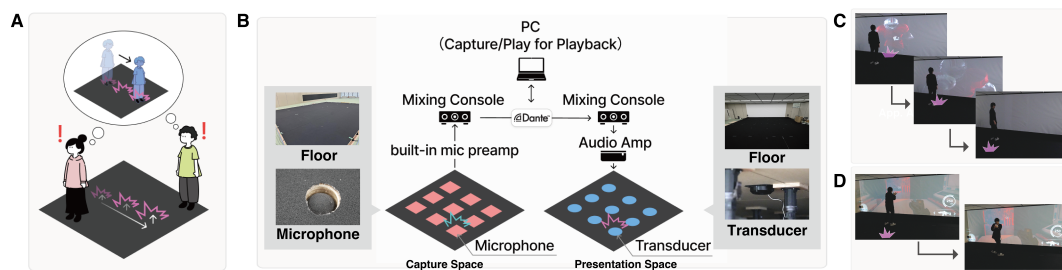


Figure 1: (A) Concept of conveying presence via vibrotactile cues, (B) System configuration and appearance, (C) Application: vibrations synced with live sports (e.g., football player running), (D) Application: sensing enemies in VR multiplayer games.

Abstract

Inspired by the human ability to sense the presence of people walking behind them through floor vibrations, we propose a concept to convey presence by transmitting footsteps and vibrotactile cues from one space to another. Based on this concept, we developed a prototype that transmits footsteps captured by multiple microphones installed on the floor surface to another space and presents a moving vibration source by switching between multiple transducers on the floor's underside. Preliminary experiments with the prototype confirmed that the direction and trajectory of the vibration source can be perceived to some extent.

CCS Concepts

• **Human-centered computing** → **Haptic devices**.

Keywords

Vibrotactile, Haptic Floors, Presence

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1 Introduction

Humans do not have eyes on their backs. However, we can sense the presence of someone walking behind us, just as we can sometimes sense someone crossing or approaching behind us while working on a PC with headphones on. These sensations may be caused not only by sight and hearing but also by floor vibrations.

Based on these experiences, we propose the concept of conveying presence by transmitting footsteps and vibrotactile cues to a remote location (Fig 1(A)). For example, there are methods such as [Orts-Escolano and et al. 2016], where the entire space is transmitted as 3D data. However, we aim to convey presence and improve the sense of immersion by presenting vibrations on the floor surface in addition to the audiovisual experience. This could be applied to various scenarios: presenting an enemy behind you in VR multiplayer games, improving presence in sports events, theatrical performances, or other live content by vibrating the floor in response to the action (Fig 1(C, D)).

Based on this concept, we developed the system shown in Fig 1(B). It captures footsteps with microphones and presents them as moving vibrations through floor transducers. Floors are common in many living spaces, providing frequent opportunities to present vibrations, and allowing many people to experience them simultaneously without the need for special equipment.

Perceiving the source of the vibration is known as epicenter localization. [Ujitoko et al. 2022] reported that the direction and

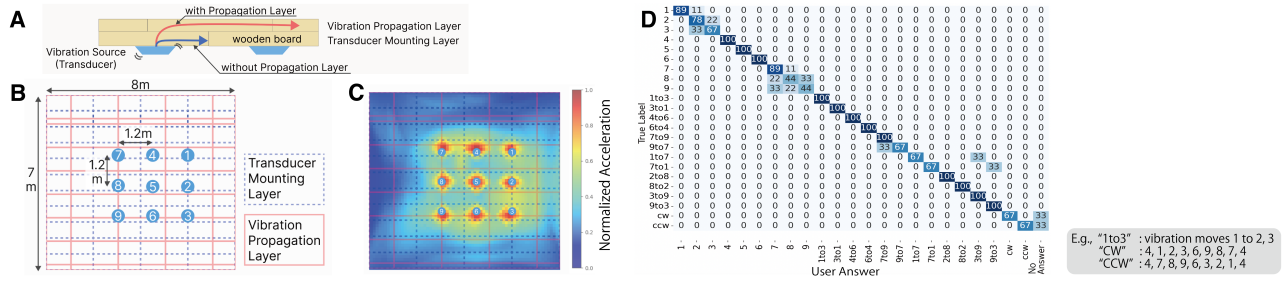


Figure 2: (A) Proposed floor structure, (B) Overhead view with transducer numbers and positions, (C) Measured vibration map, (D) Preliminary results (correlation between True Label and User Answer[%]).

trajectory of vibrations around the palm can be perceived. Similar receptors in the soles [Visieux 2020] suggest possible floor vibration localization. [Lee et al. 2017] confirmed floor vibrations enhance presence, but source localization and the trajectory of its movement have not been considered. [Takumi and et al. 2021] use sole-attached vibrators for virtual source perception, requiring special equipment or fixed positions. Our method presents a vibration field on the floor, enabling multiple users to move freely without special equipment.

2 Prototype System

An overview of the prototype system is shown in Fig 1(B).

2.1 Capture Space Configuration

Footsteps and floor vibrations are captured using nine microphones (C2, Behringer) installed in floor holes. The microphones are connected to a mixing console (QL1, Yamaha), and the 9-channel audio data is transmitted to a remote location via Dante¹, a audio transmission system. Capturing as audio allows for recording not only vibrations but also floor material and walking characteristics, using existing audio equipment for easier system setup and operation.

2.2 Presentation Space Configuration

Footsteps and floor vibrations are presented by nine transducers (Vt708, Acouve Laboratory, Inc.) installed on the underside of the floor. The transducers are connected to a mixing console (QL5, Yamaha) and controlled based on the received Dante packets. The QL5's equalizer is used to adjust the low-frequency range to make the vibrotactile sensations more perceptible.

The floor surface is constructed from standard materials, but the vibration propagation range is limited by the board size, as vibrations are damped at the end faces. To address this, vibration-propagating layers were installed in alternating directions (Fig 2(A, B)) to achieve vibration presentation beyond the board size and more natural damping. Floor vibrations were measured using a vibrometer (1332B, Showa sokki), and the results normalized to the value measured on the transducer as 1 are shown in Fig 2(C).

2.3 Preliminary Experiment

A preliminary experiment with three subjects was conducted on the localization of the vibration source (Fig 2(D)). Vibrotactile stimuli

were presented at positions 1-9, corresponding to the transducers in Fig 2(B). Subjects stood at position 5, facing direction 4, and reported the perceived location of the vibration source. A certain degree of localization was observed, but some confusion occurred in the front-back perception due to weaker vibration propagation in the front-back direction. This could be improved by modifying the floor structure and transducer placement. Additionally, some subjects felt a sense of incongruity because only vibrotactile stimuli were presented without visual cues, particularly for vibrations presented in front. Integrating visual information may improve localization and reduce this sense of incongruity. Subjects commented on the realistic sense of presence from vibrations circling around them and feeling like being crossed when vibrations moved across, confirming the effectiveness of presenting presence through vibrotactile cues.

3 Future Work

Capture Space: The current system is vulnerable to loud background noise. We'll explore environment-specific and simulation-based techniques to expand applicability and expression.

Presentation Space: Floor structure and transducer optimization will enhance directional localization. VR/AR integration will be explored for effective multimodal presentations.

Applications: Research will focus on identifying effective scenarios and technological gaps. Exploring virtual-to-physical space configurations could broaden the range of applications.

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¹<https://www.getdante.com/>